Nathan (Kaiyan) Zhang

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Research Interests

Causal inference and statistical learning for observational and high-dimensional data; Time series forecasting; Applied data science; Applications in economics, finance, labor, and text-rich data.

Education

The University of British Columbia Sep 2023 - May 2028
B.A. in Economics and Statistics; Minor in Data Science; Vancouver, BC
- Cumulative average: 93.5/100

Research Experience

Research Assistant, UBC Sauder School of Business Apr 2026 - Present
Labor, technology exposure, NLP, and empirical data construction
Supervisor: Dr. J. Frank Li and Dr. Cheng Chen
- Contribute to a faculty-led manuscript in preparation by building DuckDB and Polars pipelines linking PSID/CPS worker histories, O*NET labor metrics, and USPTO patent features through occupation crosswalks.
- Run transformer similarity workflows on UBC HPC GPU nodes to map 8.5M patent records to occupational tasks and construct lag-aware technology-exposure measures.
- Audit worker-occupation matches, parameter definitions, and panel outputs so theory-driven employment-sorting concepts become regression-ready empirical variables.

Research Assistant, UBC Sauder School of Business May 2026 - Present
Financial text, source auditing, and investment-research data
Supervisor: Dr. Allen Hu
- Build Jupyter workflows for post-2010 financial media corpora, screening major U.S. outlets for full-text access, missingness, metadata reliability, and sentiment-signal readiness. Produce a public audit artifact for downstream text-as-data work.
- Develop a reproducible workflow for downloading, processing, and auditing videos and metadata from TikTok, with the goal of assessing the platform's potential for financial sentiment and narrative research.

Research Assistant, Vancouver School of Economics at UBC Jan 2025 - May 2026
Econometrics, machine learning, and research-computing infrastructure
Supervisor: Dr. Jonathan Graves
- Developed reproducible Python and R workflows for panel econometrics, time-series forecasting, text embeddings, zero-shot classification, and computer-vision teaching modules.
- Maintained Quarto and GitHub Actions publishing pipelines for COMET and Praxis UBC materials, including online Jupyter image access for 1,000+ student users.
- Co-authored and maintained public AI literacy and research-computing materials used in UBC economics and sociology teaching.

Selected Research Projects

Narrative Innovation Response Functions In progress
Research project in progress with Jerrold Huang
Methods: financial text, event studies, semantic forecast errors, abnormal returns
- Develops an event-time design for studying market responses to unexpected components of financial narratives, using semantic forecast errors, abnormal returns, state splits, and placebo checks.

Double Machine Learning for Payment-Friction Spending Effects Nov 2025 - Dec 2025
Course research project with public report and code
Methods: Double ML, orthogonal scores, cross-fitting, CATE profiling
GitHub Report
- Designed a DoubleML workflow on a 22k person-day panel from the 2024 Atlanta Fed Diary of Consumer Payment Choice to estimate spending associations linked to low-friction payment methods while controlling for high-dimensional behavioral covariates.

Canadian Inflation Forecasting and Macro Signal Evaluation Aug 2025
Course presentation and forecasting project
Methods: VAR, time series, rolling validation, forecast benchmarking
GitHub
- Built an R and Quarto VAR forecasting pipeline for Canadian CPI and macro-financial indicators, with stationarity diagnostics, AIC lag selection, benchmark forecasts, rolling-window validation, and impulse-response interpretation.

Presentations and Workshops

Introduction to Computational Text Analysis for Research Sep 10, 2025
APSA Migration and Citizenship Pre-Conference; Workshop materials contributor and technical facilitator Workshop page

- Contributed demo, slides, and training materials for a hands-on computational text analysis workshop and assisted with in-session technical support.

Teaching and Research Computing Materials

ECON 227 LLM Prediction Module
Co-author Module Link

- Co-authored course-adopted materials connecting next-word prediction, token probabilities, and economics examples for students learning LLMs through applied research.

SOCI 280 Research Computing Materials
Research support, planning, editorial review, and maintenance Module Link

- Supported material collection, planning, review, and maintenance for computational social science teaching materials.

Praxis UBC AI Literacy Modules
Contributor and maintainer Website Link

- Contributed public teaching modules on LLMs, embeddings, CNNs, convolutions, and research computing for non-CS student audiences.

Teaching Experience

Teaching Assistant, UBC Chinese Language Program 2024 - 2025
Vancouver, BC

- Supported CHIN 131 and CHIN 133 through grading, classroom support, volunteer training, and feedback coordination.

Awards and Scholarships

Work Learn International Undergraduate Research Awards (WLIURA) (UBC) 2026
Economics Undergraduate Scholarship (Vancouver School of Economics, UBC) 2025
Faculty of Arts International Student Scholarship (UBC Faculty of Arts) 2024, 2025

Selected Coursework

Statistical theory and methods: STAT 302 - Introduction to Probability (91); STAT 305 - Introduction to Statistical Inference (100); STAT 406 - Methods for Statistical Learning (97); STAT 443 - Time Series and Forecasting (95)

Mathematics and optimization: MATH 220 - Mathematical Proof (100); MATH 307 - Applied Linear Algebra (100); CPSC 406 - Computational Optimization (92)

Econometrics, causal inference, and machine learning: ECON 398 - Introduction to Causal Methods (98); ECON 425 - Advanced Econometrics (98); CPSC 340 - Machine Learning and Data Mining (93)

Economics Theory: ECON 304 - Honours Intermediate Microeconomics I (89); ECON 305 - Honours Intermediate Macroeconomics I (91); ECON 307 - Honours Intermediate Macroeconomics II (96)

Registered coursework: MATH 320 - Real Variables I; STAT 460 - Statistical Inference I; STAT 461 - Statistical Inference II; MATH 303 - Introduction to Stochastic Processes; CPSC 440 - Advanced Machine Learning; AI 360 - Deep Learning; CPSC 423 - Natural Language Processing; ECON 421 - Introduction to Game Theory and Applications

Technical Skills

Statistical computing: Python, R, SQL, Julia, MATLAB, Jupyter, Quarto, Git/GitHub, HPC/Slurm

Data and ML libraries: pandas, NumPy, scikit-learn, PyTorch, Transformers, DuckDB, Polars, tidyverse, fixest, statsmodels

Methods: causal inference, statistical learning, time series forecasting, econometric modeling, text-as-data, reproducible workflows

Languages

Mandarin Chinese: native; English: academic working proficiency; Japanese: JLPT N3 certified; currently N2 level proficiency